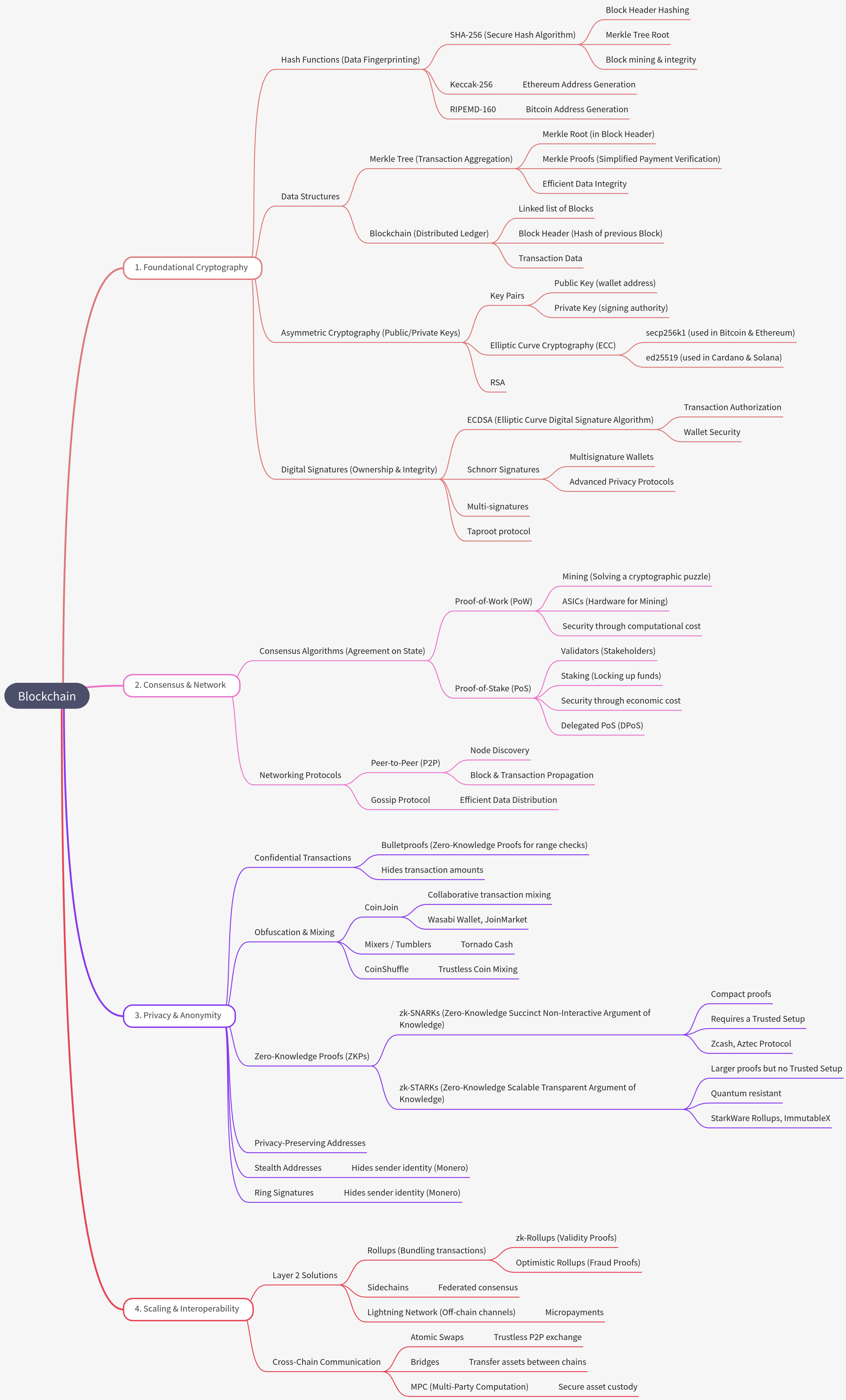
Blockchain
1. Foundational Cryptography
Hash Functions (Data Fingerprinting)
SHA-256 (Secure Hash Algorithm)
Block Header Hashing
Merkle Tree Root
Block mining & integrity
Keccak-256
Ethereum Address Generation
RIPEMD-160
Bitcoin Address Generation
Data Structures
Merkle Tree (Transaction Aggregation)
Merkle Root (in Block Header)
Merkle Proofs (Simplified Payment Verification)
Efficient Data Integrity
Blockchain (Distributed Ledger)
Linked list of Blocks
Block Header (Hash of previous Block)
Transaction Data
Asymmetric Cryptography (Public/Private Keys)
Key Pairs
Public Key (wallet address)
Private Key (signing authority)
Elliptic Curve Cryptography (ECC)
secp256k1 (used in Bitcoin & Ethereum)
ed25519 (used in Cardano & Solana)
RSA
Digital Signatures (Ownership & Integrity)
ECDSA (Elliptic Curve Digital Signature Algorithm)
Transaction Authorization
Wallet Security
Schnorr Signatures
Multisignature Wallets
Advanced Privacy Protocols
Multi-signatures
Taproot protocol
2. Consensus & Network
Consensus Algorithms (Agreement on State)
Proof-of-Work (PoW)
Mining (Solving a cryptographic puzzle)
ASICs (Hardware for Mining)
Security through computational cost
Proof-of-Stake (PoS)
Validators (Stakeholders)
Staking (Locking up funds)
Security through economic cost
Delegated PoS (DPoS)
Networking Protocols
Peer-to-Peer (P2P)
Node Discovery
Block & Transaction Propagation
Gossip Protocol
Efficient Data Distribution
3. Privacy & Anonymity
Confidential Transactions
Bulletproofs (Zero-Knowledge Proofs for range checks)
Hides transaction amounts
Obfuscation & Mixing
CoinJoin
Collaborative transaction mixing
Wasabi Wallet, JoinMarket
Mixers / Tumblers
Tornado Cash
CoinShuffle
Trustless Coin Mixing
Zero-Knowledge Proofs (ZKPs)
zk-SNARKs (Zero-Knowledge Succinct Non-Interactive Argument of Knowledge)
Compact proofs
Requires a Trusted Setup
Zcash, Aztec Protocol
zk-STARKs (Zero-Knowledge Scalable Transparent Argument of Knowledge)
Larger proofs but no Trusted Setup
Quantum resistant
StarkWare Rollups, ImmutableX
Privacy-Preserving Addresses
Stealth Addresses
Hides sender identity (Monero)
Ring Signatures
Hides sender identity (Monero)
4. Scaling & Interoperability
Layer 2 Solutions
Rollups (Bundling transactions)
zk-Rollups (Validity Proofs)
Optimistic Rollups (Fraud Proofs)
Sidechains
Federated consensus
Lightning Network (Off-chain channels)
Micropayments
Cross-Chain Communication
Atomic Swaps
Trustless P2P exchange
Bridges
Transfer assets between chains
MPC (Multi-Party Computation)
Secure asset custody